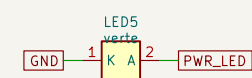
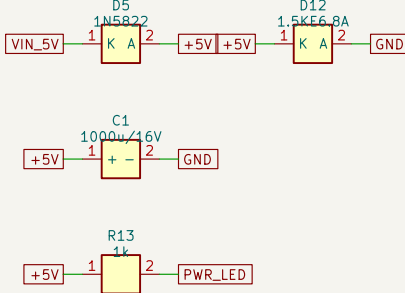
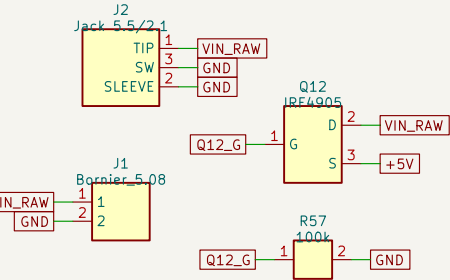
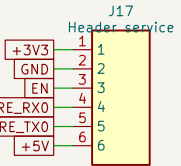


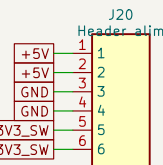
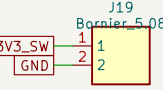
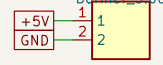
ALIMENTATION 5V (3A recommandé) – jack OU bornier J1 OU sortie buck J37, tous sur VIN_RAW.
Q12 (IRF4905, VGS=–5V) = anti-inversion : un fil inversé ne détruit plus la carte. D12 = TVS 6.8V.
D5 protège le rail si l'USB du DevKit est branché en même temps.



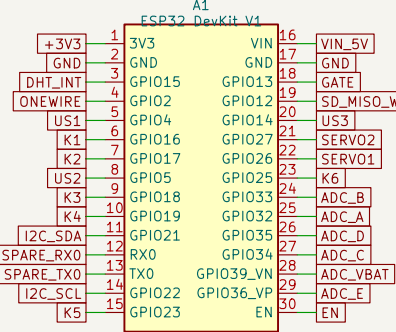
HEADER SERVICE J17 : 3V3 / GND / EN / RX0 / TX0 / +5V uniquement.
Tous les autres GPIO sont consommés par la carte universelle (GPIO36 = ADC_E, GPIO39 = ADC_VBAT). Laisser RX0/TX0 libres pendant le flash USB.



Distribution 5V / 3V3=capteurs / GND : borniers à vis + rail header. Points de test TP1..TP4.



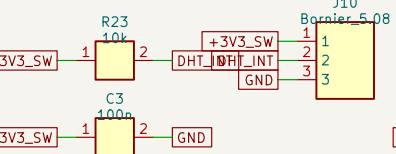
SITE A1 : ESP32 DevKit V1 (30 broches, socketé) – UN SEUL module peuplé (A1 OU A2).
Emprise à DEUX entraxes (25.4 = rangée A, 27.94 = rangée A') : souder le support droit là où tombent les broches du module.
Env. carte universelle : msp/n3pp esp32dev_universal_test, ffp5cs wroom–universal–test.
Sur une unité ffp5cs, ÔTER JP1 avant un flash UART (le pull–up R24 sur GPIO2 bloque le bootloader).



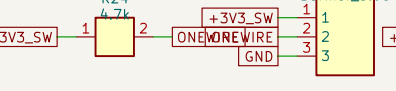
SERVO 5V (nets SERV01/SERV02) – WROOM : GPIO26/27, S3 : IO21/IO47.
Signal en série R20/R21 (220R). Alimentation prise directement sur le rail +5V.



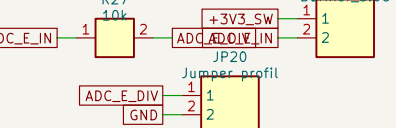
AIR : DHT11 (option DHT22, –DUSE-DHT22) – pull–up 10k.



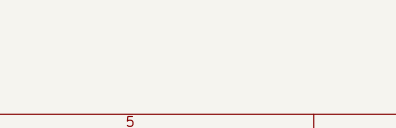
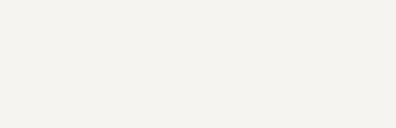
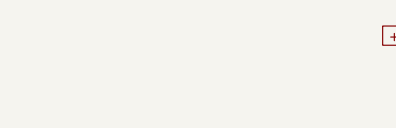
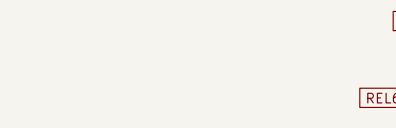
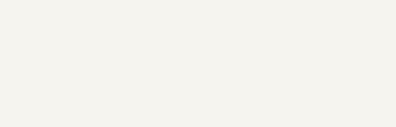
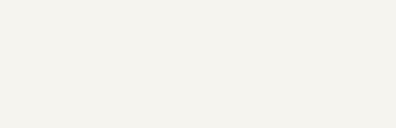
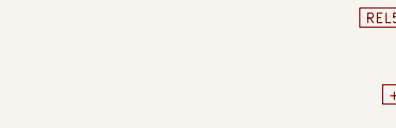
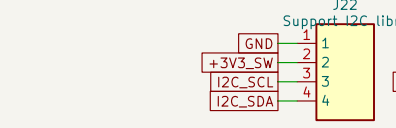
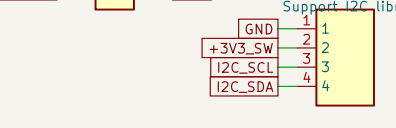
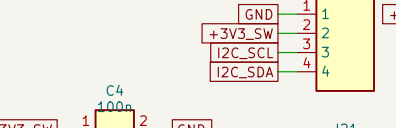
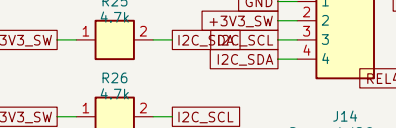
EAU : DS18B20 (1–Wire, pull–up 4.7k).



LUMIERE : LDR déportée sur J12, net ADC_E (GPIO36) – 10k 533.
Bas de pont R27 10k en circuit si JP20 FERME (LDR n3pp/ffp5cs) : OUVERT pour un module AO (msp/n3pp/ffp5cs).
Filtre série 1k + 100n sur chaque entrée ADC (fils longs).



I2C : OLED SSD1306 0x3C + 3 supports extension (DS3231, INA226...).



PONT DIVISEUR VBAT PERMANENT (rev 0.2, plus de 35200) : R36 100k + JP12 : 1–2 = 22kV, 2–3 = 100k (15 Li–ion, msp/n3pp). R49/C13 filtre 5V3 à 10amp 3V3.

